

FROM GENERATIVE CAPABILITY TO GOVERNED INSTITUTIONAL DELIBERATION

FROM STATUTE TO REFORM: A GOVERNED 50-AGENT SWARM FOR MEXICAN SECURITIES LAW

A Colab Laboratory for Diagnosis, Adversarial Review, Selection, Legislative Drafting and Human Authorization

A pedagogical research monograph based on a synthetic fifty-agent experiment that studies the Mexican Ley del Mercado de Valores, organizes institutional disagreement, constrains model authority through an explicit harness, selects exactly twenty candidate articles, drafts a preliminary reform package, and preserves the final act of legal judgment for human review.

CENTRAL PROPOSITION

The central object is not a powerful legal chatbot. It is a governed deliberative system. Model intelligence is distributed across specialized synthetic roles, but source authority, proposal identity, scoring, selection constraints, validation and final authorization remain external to the model. The experiment therefore asks a deeper question than whether an LLM can draft legislation: can artificial intelligence be organized as an auditable institution for structured disagreement?

Executive overview: the problem is institutional design, not text generation

The notebook underlying this monograph begins from a deliberately demanding premise. A reform of securities law should not be reduced to a single prompt asking a language model to “improve” a statute. The difficult part is not the production of fluent legal language. The difficult part is the institutional work that precedes and follows drafting: identifying real frictions, preserving the authority of the legal corpus, representing competing interests, exposing trade-offs, challenging proposals, deciding which provisions deserve attention, controlling who may change what, and preserving a record that allows a third party to reconstruct the path from diagnosis to proposed amendment.

The experiment therefore treats the Mexican *Ley del Mercado de Valores* (LMV) as the object of a governed multi-agent process. Fifty synthetic agents are assigned to eight institutional families: practicing lawyers; legislative and public-policy specialists; regulators; issuers; exchanges and market infrastructure; broker-dealers; institutional investors; and comparative-law and academic specialists. They use the same underlying language model, but they do not receive the same mandate. Each family is given a different lens, declared bias and criterion of success. Diversity is created by institutional position rather than by pretending that identical model instances become different simply because they receive different names.

The system then separates functions. The language model is used where linguistic and interpretive judgment is useful: diagnosis, policy framing, proposal generation, adversarial criticism, drafting and integration. Python is used where deterministic control is preferable: counting agents, segmenting the corpus, preserving article identifiers, checking whether cited provisions exist, ranking proposals, enforcing the twenty-article constraint, verifying reviewer independence, and generating an audit trail. The human remains outside both layers as the source of authorization.

The stored execution is especially instructive because it does not end in artificial triumphalism. The run reports 50 configured agents, 176 diagnostic findings from 37 contributing agents, 169 distinct diagnosed LMV articles, 33 proposals, 33 adversarial reviews, exactly 20 selected articles and 20 completed drafts. Yet the drafting stage also reports that only one of the twenty article drafts received a positive legislative-technique approval, while nineteen remained flagged for revision. The final validation report nevertheless returned `ready_for_human_review = true` because no deterministic integrity error remained. That distinction is central: *ready for human review* does not mean *ready for Congress*.

THE EXPERIMENT IN ONE VIEW

Configured synthetic agents	50
Diagnostic findings in the stored run	176
Agents contributing usable diagnoses	37
Distinct diagnosed LMV articles	169
Preliminary reform proposals	33
Independent adversarial reviews	33
Distinct articles selected	20
Article drafts completed	20
Drafts positively approved by technique review	1
Drafts still flagged by technique review	19
Final deterministic status	Ready for human legal review

This result reveals the governing idea of the laboratory. A capable model may produce a plausible reform package. A governed system must also know how to preserve objections, expose incompleteness, distinguish mechanical integrity from legal validity, and stop before converting synthetic deliberation into public authority.

Research question: can a swarm behave like a deliberative institution?

The research question is not whether fifty agents outperform one agent by brute force. It is whether the architecture can turn general-purpose model intelligence into a process with recognizable institutional properties. Those properties include specialization, source discipline, separation of functions, structured dissent, review by counter-perspectives, bounded authority, reproducibility and explicit escalation to human judgment.

The LMV is a useful laboratory because securities regulation sits at the intersection of several objectives that cannot be optimized independently. Market access can conflict with compliance cost. Investor protection can conflict with speed or product flexibility. Supervisory discretion can support resilience while generating uncertainty. Disclosure can improve price formation while creating burdens that are not equally material for every participant. Market infrastructure, governance, intermediation, enforcement and innovation interact across the statute. A proposal that appears attractive from one institutional vantage point can produce costs elsewhere.

A single conversational model tends to compress those conflicts into one voice. The swarm instead attempts to preserve them. The practicing lawyer asks whether a rule is coherent and enforceable. The regulator asks whether it is supervisable and robust. The issuer asks whether the cost is proportionate. The institutional investor asks whether governance, disclosure and minority protection remain credible. Exchanges and infrastructure agents focus on liquidity, listing and market functioning. Broker-dealers focus on placement, suitability and liability. Comparative-law agents are instructed to identify functional lessons without treating foreign rules as Mexican law.

DESIGN PRINCIPLE: DISAGREEMENT IS DATA

The experiment does not treat consensus as the objective function. A useful legislative process should retain material disagreement long enough for it to inform the final decision. The system therefore records supporting roles, opposing roles, unresolved questions, strongest objections and implementation risks. A proposal is stronger when it survives criticism, not when the architecture suppresses criticism.

The notebook expresses this idea through an inner loop and an outer loop. The inner loop performs substantive work: diagnose, propose, criticize, rank and draft. The outer loop evaluates the process itself: whether participation is concentrated, whether some institutional family has disappeared, whether proposals duplicate one another, whether the corpus supports the claims, and whether the apparent consensus may be a cascade rather than independent agreement. This second-order layer is what turns a collection of model calls into something closer to institutional machinery.

Foundations: environment, source authority and article-level memory

The first three computational layers establish the conditions under which later reasoning is allowed to occur. They are easy to underestimate because they do not look like legal analysis. In fact, they define the epistemic constitution of the experiment.

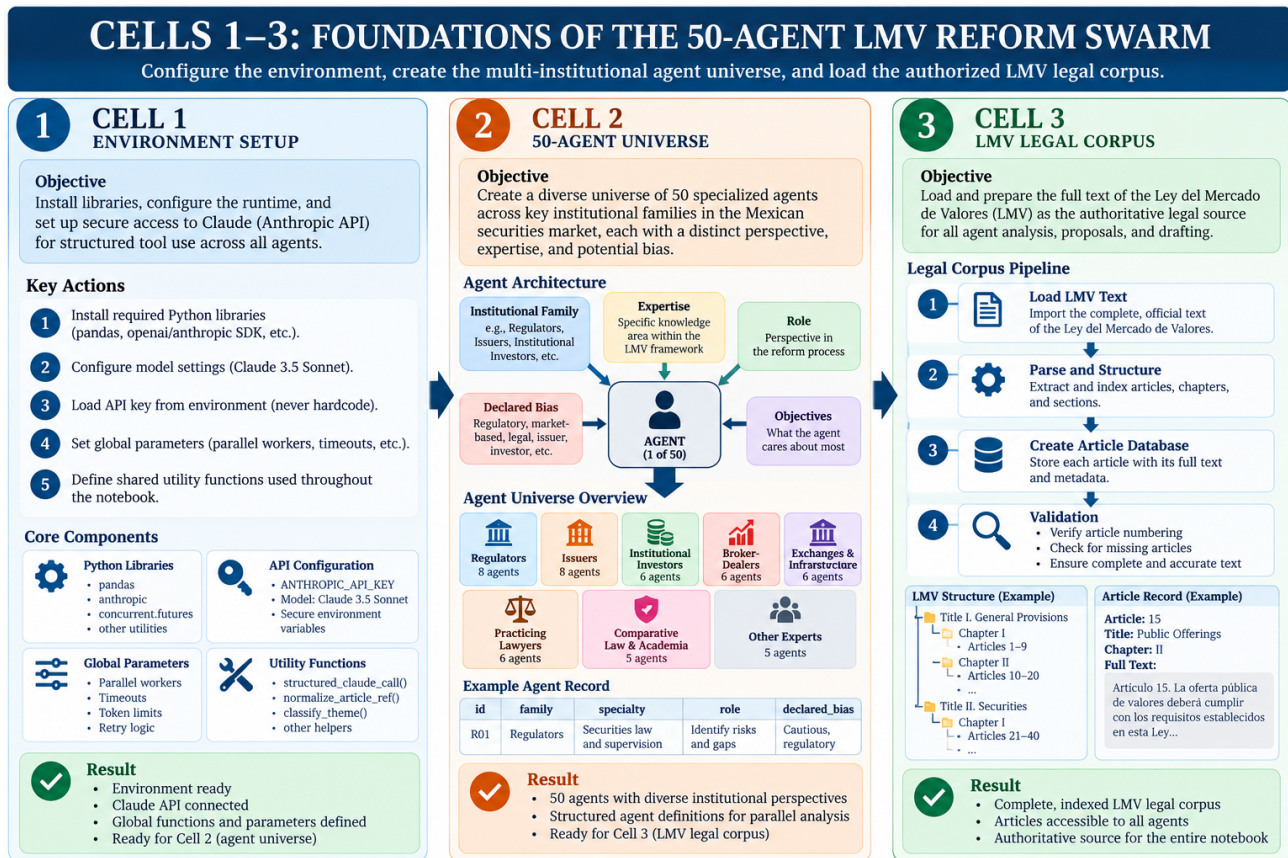


Figure 1: Cells 1–3: runtime configuration, construction of the fifty-agent universe, and ingestion and segmentation of the authorized LMV corpus.

Cell 1: reproducibility, dependencies and execution governance

The configuration cell declares dependencies, fixes a random seed, specifies the model identifier and centralizes API access. Credentials are retrieved through Colab secrets and are neither printed nor written into notebook outputs. A `LIVE_LLM` switch allows the pipeline to run in a demonstration mode without silently replacing the declared model or improvising credentials.

This is a governance decision rather than a convenience. When model access is centralized, every agent is subject to the same token limits, retry policy, output contract and audit mechanism. The system avoids uncontrolled per-agent settings and does not rely on `temperature=0` as a substitute for governance. Variation is accepted where argument diversity is useful; determinism is reintroduced through structured output and validation.

Cell 1 should therefore be read as the notebook’s execution charter. It identifies the software libraries on which later cells depend, initializes the structures that will hold records, and fixes the settings that must remain stable if a reviewer is to understand what happened. This work is not ancillary to the legal experiment. A multi-agent pipeline can produce hundreds of intermediate objects, and small configuration differences can change which objects are created, how failures are retried and which outputs survive. By concentrating those choices in one visible place, the notebook gives the operator a practical inventory of assumptions before any legal interpretation begins. The declared random seed does not make a language model fully deterministic, but it stabilizes notebook-controlled operations that use randomized assignment or ordering. The model identifier similarly prevents an unnoticed substitution from being mistaken for a replication of the same experiment.

The cell also establishes credential discipline. An API key is treated as a runtime secret rather than as research content. It is obtained from the notebook environment, checked for availability and withheld from printed output, exported files and prompts. This separation protects the operator, but it also improves methodological clarity: a third party may inspect the notebook without receiving authority to spend against the original user’s account. The `LIVE_LLM` control makes another boundary explicit. When live access is disabled, the notebook can demonstrate

control flow or consume stored material, but it must not imply that fresh model deliberation occurred. Conversely, when live access is enabled, each request passes through the declared gateway and becomes part of the audit record. The switch therefore distinguishes pedagogical or offline execution from model-mediated execution instead of allowing the two modes to blur.

Central configuration is especially important in a fifty-agent setting because apparent plurality can conceal infrastructural inconsistency. If individual agents were allowed to choose their own models, token budgets, retry rules or response formats, differences in output could be caused by hidden technical variation rather than institutional perspective. Cell 1 removes much of that ambiguity. It defines common operating conditions while leaving later prompts free to express different roles. In experimental terms, the infrastructure is held comparatively constant so that mandate becomes the principal designed source of variation. In governance terms, no synthetic participant receives a private channel to a model with different privileges.

Failure behavior is part of the same charter. A transient request error, malformed structured response or missing secret should become a recorded event, not a reason for the notebook to silently fabricate a substitute. The cell's configuration enables bounded retries and status reporting, allowing subsequent stages to distinguish an agent that returned no usable answer from an agent that was never called. That distinction matters when the governance layer later measures participation. A missing contribution may reveal a model failure, a validation rejection or an assignment problem; it should not automatically be interpreted as institutional silence.

Finally, Cell 1 defines what reproducibility can and cannot mean for this project. Exact prose may vary across live language-model calls even when the notebook is unchanged. The reproducible object is therefore the governed procedure: the same declared model class, source boundary, schemas, constraints, logging rules and sequence of gates. A future rerun can compare results meaningfully because the conditions are recorded, not because the experiment promises byte-for-byte identity. This modest definition is more credible than claiming determinism where none exists. It also anticipates the monograph's central theme: reliable institutional use of generative models depends less on eliminating all variation than on making variation observable, bounded and reviewable.

Cell 2: the authorized corpus as a boundary of legal authority

The second foundation is the corpus. The notebook treats the LMV supplied by the user or obtained from an official source as the primary legal authority. Additional materials can be loaded, but they are labeled with provenance and jurisdiction. Comparative material may widen the space of alternatives; it is not allowed to masquerade as binding Mexican law.

Each source receives an identifier and hash. The ingestion routine checks that enough extractable text exists before the pipeline proceeds. This matters because one of the most dangerous failure modes in legal AI is confident reasoning over a document the system did not actually parse. The notebook's source harness makes a different claim: *if the legal text is not in the authorized corpus, the model does not receive authority to invent it.*

Cell 2 operationalizes that claim by converting source acquisition into an explicit authorization event. The user supplies or designates the LMV text, and the notebook records where it came from, what jurisdiction it belongs to and how it should be treated. This is a crucial difference between retrieval and authority. A document can be technically available to the system without being legally authoritative for the task. Comparative legislation, commentary, prior versions and secondary regulation may all be informative, but they do not have the same status as the designated Mexican statute. The cell therefore builds a corpus whose entries carry metadata rather than placing all text into an undifferentiated prompt. Later agents can be told not only what a passage says but also why the system considers that passage usable.

The source identifier and hash support two forms of traceability. First, they let the notebook refer to a stable corpus item throughout the run. Second, they allow a reviewer to detect whether the underlying file has changed between executions. A hash does not prove that a source is official, current or correctly interpreted; those remain substantive questions for human review. It does, however, prove that two recorded artifacts are or are not byte-identical. In a legislative workflow, that modest property is valuable. A proposal generated from one version of a statute should not be presented as though it were generated from another, and a later correction to the source should be visible as a new corpus state.

Text extraction is treated as a validation problem rather than as a clerical step. PDFs may contain scanned pages, broken character maps, repeated headers or missing text. A model given an empty or corrupted extraction can still produce persuasive legal language from general patterns, making the failure difficult to notice. Cell 2 counters this risk by requiring a minimum amount of recoverable text and by preserving provenance alongside the extraction. The check is intentionally mechanical: it does not certify completeness, but it prevents the most obvious condition in which the notebook claims to analyze a statute that it never successfully read. An institutional implementation would extend the same idea with page-level coverage reports, optical-character-recognition confidence and human spot checks.

The cell also establishes a hierarchy for supplementary sources. Official Mexican legal materials can be admitted with

an appropriate label. Nonbinding guidance can be marked as interpretive rather than statutory. Foreign regimes can be marked as comparative. Academic or market evidence can be added as analysis, not law. These labels give later prompts a disciplined vocabulary for uncertainty. An agent may say that a foreign mechanism suggests an option, but it may not cite that option as an existing LMV obligation. Similarly, it may identify a factual assumption that requires market data without converting an unsupported assumption into a legal finding.

This boundary reduces, but does not eliminate, hallucination. The model may still misread an authorized passage, omit an exception or overstate the relationship between provisions. The design response is not to pretend that retrieval guarantees correctness. It is to ensure that every claim capable of being checked has a recoverable source context and that missing authority becomes visible. The cell therefore turns epistemic humility into system behavior. When relevant material is absent, the appropriate output is uncertainty or a request for supplementation, not invented certainty.

For the rest of the notebook, Cell 2 functions like a constitution for evidence. Diagnoses, proposals, reviews and drafts inherit the corpus boundary; none of those later stages may enlarge it merely because a fluent answer would benefit from additional material. Only the human operator can authorize a new source class or a replacement text. That design keeps source power outside the synthetic agents and makes corpus adequacy a reviewable human decision. In a domain where the difference between binding text, administrative practice and comparative analogy is legally consequential, this separation is foundational rather than cosmetic.

Cell 3: article-level representation and deterministic legal identity

The third step segments the statute into article-level records. This creates a lightweight legal memory that supports targeted retrieval, validation and traceability. The article number is not merely a string generated by the model; it is an identifier checked against a dictionary created from the corpus.

The representation is intentionally simpler than a production-grade legal parser. Articles Bis or Ter, transitory provisions, nested fractions and PDF irregularities may require more sophisticated treatment. The pedagogical value lies precisely in making the approximation visible. The system declares its unit of analysis, checks coverage and allows the human reviewer to see when extraction looks abnormal.

Cell 3 transforms the authorized statute from a long stream of text into a set of addressable legal records. Each record links an article label to the extracted language and to the source from which it came. This enables later cells to retrieve a bounded provision instead of repeatedly sending the whole statute to the model. The resulting economy is computational, but the more important benefit is conceptual: every diagnosis and proposal can be attached to a legal unit whose existence is established outside the model. Article identity becomes a property of the parsed corpus rather than a guess produced in prose.

The segmentation routine is deliberately deterministic. Given the same extracted text and parsing rules, it should identify the same candidate boundaries and normalize labels in the same way. This permits validation at several points. When an agent cites Article 217, the notebook can test whether that label appears in the article dictionary. When selection chooses twenty provisions, the notebook can confirm that all twenty belong to the authorized statute. When drafting finishes, it can compare the drafted set with the selected set. None of those checks requires the language model to judge its own compliance. The parser thus supplies a small but essential piece of institutional memory.

Normalization deserves particular attention because legal identifiers are not always simple integers. Mexican statutes may contain provisions labeled Bis, Ter or with more complex extensions; spacing, capitalization and extraction artifacts can produce multiple strings for what a human recognizes as the same article. Cell 3 must preserve the legal label while also creating a consistent comparison key. If normalization is too aggressive, distinct provisions may collapse. If it is too weak, one provision may appear under several forms and defeat deduplication. The notebook's lightweight method is suitable for the pedagogical corpus, but the monograph openly treats edge cases as a limitation requiring specialist review in a production deployment.

Article-level records also create a basis for targeted context assembly. An agent assigned to a particular subject can receive the relevant provision and, where the notebook supports it, neighboring or cross-referenced material. This reduces prompt noise and makes it easier to inspect what evidence was actually available for a finding. Yet targeted retrieval introduces its own risk: a provision can be misunderstood when separated from definitions, exceptions or institutional chapters elsewhere in the statute. The cell therefore should not be read as claiming that an article is legally self-contained. It provides an indexing unit, not a theory of statutory interpretation. Dependencies identified later must be recorded and escalated rather than silently assumed away.

The representation also supports coverage analysis. The notebook can count how many distinct articles received diagnoses, which provisions attracted proposals and whether the final portfolio concentrates in a narrow region of the statute. These measures help the outer loop distinguish broad participation from repeated attention to a small set of familiar issues. Coverage is not itself quality: mentioning many articles can produce superficial analysis. Nevertheless, without an article index the system could not even observe the distribution of attention, and governance would

depend on impressions rather than records.

Most importantly, Cell 3 prevents later agents from acquiring the power to create legal objects through assertion. A model can recommend that a provision be studied, but if the label does not resolve against the corpus the finding is quarantined or rejected. A model can draft replacement language, but it cannot use that language to retroactively establish that the referenced article existed. This asymmetry is intentional. Interpretation remains generative; identity remains deterministic. The cell thereby illustrates the architecture's broader method: assign flexible language tasks to the model, reserve crisp membership and integrity questions for code, and expose uncertain boundaries to humans instead of concealing them behind fluent output.

WHY THIS MATTERS

A language model can produce highly plausible references to non-existent provisions. The notebook responds by making article identity a deterministic object. Later stages may interpret an article, criticize a proposal or draft language, but they do not get to redefine whether the source article exists.

Cell 4: the 50-agent institution—specialization without simulated authority

The swarm is constructed as eight institutional families with explicit counts: eight practicing lawyers; six legislative and policy specialists; seven regulators; seven issuer representatives; five exchange and infrastructure specialists; six broker-dealer specialists; six institutional investors; and five comparative-law and academic specialists. All start with equal numerical weight.

The agents are intentionally functional rather than biographical. They do not impersonate real individuals. That design avoids falsely attributing views to identifiable practitioners or institutions and makes the experiment's status clear: these are synthetic institutional perspectives, not a consultation exercise.

Each agent receives five core attributes: role, specialty, objective, declared bias and criterion of success. The declared bias is important. The architecture does not attempt to eliminate institutional incentives; it makes them inspectable. An issuer agent may prefer lower justified compliance frictions. An investor agent may prefer stronger disclosure and governance. A regulator may prefer rules that are easier to supervise and less vulnerable to market abuse. The model is therefore asked to reason *from a position*, while the surrounding system prevents any one position from becoming self-authenticating.

EIGHT FAMILIES, ONE MODEL

The experiment does not claim model diversity. All agents use the same underlying model. What varies is institutional mandate. This is enough to create structured perspective diversity, but not enough to eliminate correlated model error. The limitation becomes important later when interpreting apparent consensus.

A production system could extend diversity in at least three directions. First, it could use multiple model families rather than one. Second, it could add verified human submissions as a separate evidence layer instead of turning them into synthetic personas. Third, it could represent expertise more granularly—for example, separating public offerings, market abuse, CCP governance, custody, enforcement, corporate law and constitutional review. The notebook deliberately stops earlier because its objective is to make the basic architecture inspectable.

Cell 4 is the point at which a single technical system becomes an institutional simulation. It constructs fifty records, assigns every record a stable identifier and associates it with one of the eight families. The numerical allocation is not presented as an empirical claim about how a real legislative committee should be composed. It is a design choice that creates enough participants for repeated assignments while preventing any one family from constituting a majority. Because the counts are explicit, a reader can challenge them, rerun the experiment with another distribution or test whether results change when underrepresented perspectives receive more places. The allocation is therefore a transparent parameter rather than a hidden assumption.

The distinction between a role and a person is essential. An agent labeled as a regulator does not possess public office, access to confidential supervisory information or the authority of an agency. It receives a prompt that asks it to emphasize supervisability, enforcement risk and market integrity. Likewise, an issuer agent is not evidence of what Mexican issuers believe. It is a structured invitation to examine compliance cost, capital access and operational feasibility. This careful framing limits anthropomorphic overstatement. The agents are analytical instruments used to generate arguments from declared positions; they are not stakeholders whose consent can legitimate a reform.

Declared bias performs two functions. Substantively, it encourages the model to notice consequences that a generic prompt may smooth over. Procedurally, it lets reviewers interpret a contribution in context. A strong objection from an investor-protection lens and a strong objection from an issuer-cost lens do not cancel each other mechanically; they reveal a trade-off that the eventual human decision must address. The criterion of success provides a similar anchor. It tells an agent what a satisfactory outcome would look like without giving it power to decide whether that outcome governs the whole system. In this sense, Cell 4 distributes attention, not authority.

Stable agent identifiers also make participation measurable. Later cells can record which agents submitted valid diagnoses, led proposals, performed reviews and drafted text. Without those identifiers, the phrase “fifty-agent swarm” would be largely rhetorical. With them, the outer loop can detect that only 37 agents produced usable diagnoses in the stored execution, examine family concentration and balance review loads. The design therefore distinguishes configured membership from realized influence. Merely creating a persona is not evidence that its perspective affected the outcome.

The cell's deepest limitation is correlated reasoning. All fifty records ultimately invoke the same model and inherit many of the same linguistic habits, background priors and failure modes. Role prompts can redirect attention, but they cannot create fifty independent minds. Apparent convergence must consequently be treated as a signal generated by the architecture, not as a poll of independent experts. The notebook addresses this limitation through adversarial assignment, deterministic constraints and human review, but it cannot remove it. By stating the limitation next to the construction of the swarm, the monograph keeps institutional diversity and model diversity conceptually

separate.

Cell 4 thus supplies a reusable pattern: represent relevant viewpoints explicitly, disclose the lens assigned to each, preserve the identity of contributions and prohibit synthetic roles from claiming the status of real participants. A production process could place genuine submissions beside these synthetic analyses, clearly labeled and governed by different evidentiary rules. The pedagogical notebook stops before that step. Its achievement is narrower but important: it makes disagreement an architectural feature and ensures that the resulting plurality remains inspectable rather than theatrical.

Cell 5: the harness—bounded model judgment under deterministic control

Cell 5 creates the single gateway through which the agents interact with the model. This is the notebook's most important control surface. The common system rules tell every agent that Mexican law is primary, sources may not be invented, uncertainty must be declared, and diagnosis is not authorization. The agent-specific identity then adds the relevant institutional lens.

Outputs are not accepted as free-form prose when later computation depends on them. The notebook uses structured schemas with required fields. Diagnosis requires problem statements, article numbers, evidence, policy objectives, beneficiaries, risks and priority. Adversarial review requires eight numerical dimensions, strongest objection, unintended consequences, missing dependencies, comparative-source status and a KEEP/REVISE/DROP recommendation. Drafting requires proposed legal text, explanation, cross-references and follow-up dependencies.

That architecture creates a useful separation of capabilities:

- the model supplies interpretation, argument and language;
- Python preserves identity, validates shape and applies constraints;
- the corpus supplies legal source authority;
- the human supplies final authorization.

AUTHORITY MODEL

The model may propose. It may diagnose a friction, frame a policy objective, draft language or identify an objection.

The model may evaluate. It may score a proposal under a specified rubric when it did not author that proposal.

The model may not authorize. Article identity, the twenty-article cap, validation rules and the decision to proceed remain outside the agent that generated the recommendation.

The audit log records agent, task, prompt hash, model and execution status. This means the final document can be accompanied by a process record rather than presented as a detached block of generated prose. In legal and regulated environments, provenance is not merely debugging metadata. It is part of accountability.

Cell 5 implements the operational boundary between the synthetic institution and the language model. Rather than allowing later cells to call the model directly, the gateway receives a task, an agent identity, authorized context and an expected output schema. It assembles those components in a consistent order, sends the request, captures the response and records whether validation succeeded. Centralization makes common rules genuinely common. A later drafting cell cannot quietly omit the source warning, and an individual agent cannot grant itself a larger context window or a different output contract simply because its task is difficult.

The structured schemas are not merely formatting aids. They define the minimum record required for the next institutional stage. A diagnosis that lacks an article reference or evidence field cannot be treated as equivalent to one that can be traced. A review without a strongest objection cannot satisfy the adversarial function even if its prose sounds sophisticated. A draft without dependencies cannot demonstrate that cross-references were considered. Schema validation therefore converts procedural expectations into machine-checkable conditions. It does not determine whether the content is legally sound, but it stops missing components from disappearing inside elegant paragraphs.

The gateway also provides a disciplined response to malformed output. Generative models sometimes add commentary around structured data, omit a required field or return an unexpected value. The notebook may retry within declared limits or mark the task as unsuccessful, but it should not silently rewrite substantive content in order to force validation. This preserves a distinction between transport repair and legal judgment. Removing a stray code fence may be a mechanical normalization; inventing a missing legal rationale is not. Recorded failure is preferable to an apparently complete artifact whose missing reasoning was supplied by invisible code.

Prompt hashes support comparison without requiring the audit file to expose every full prompt in every view. If a task is rerun after the prompt changes, the hash changes and the execution can be distinguished. Together with the agent identifier, task name, model and status, this creates a chain of provenance for each artifact. A production environment would add timestamps, versioned schemas, token usage, latency, request identifiers and access controls. The notebook's simpler log nonetheless demonstrates the governing principle: generated text should arrive with enough metadata to reconstruct how it entered the process.

Cell 5 also prevents role instructions from overriding system rules. The regulator's objective, the issuer's bias or

the comparative scholar's interest in foreign models remains subordinate to the shared requirements concerning source authority, uncertainty and bounded action. This hierarchy matters because role-play prompts can otherwise encourage confident advocacy. The harness tells the model that inhabiting a perspective is a method for analysis, not permission to fabricate evidence or claim decision rights. Institutional identity is nested inside a common legal and procedural frame.

The resulting division of labor is deliberate. Language generation remains flexible where the task demands interpretation, criticism or drafting. Code remains strict where the task concerns membership, required fields, counts and status transitions. Human authority remains necessary where a choice has legal, constitutional or political consequence. Cell 5 is where these layers meet, and its success should be judged by whether it preserves their boundaries. A more fluent gateway would not be better if it obscured failure, weakened schemas or allowed prompts to expand their own authority. The point of the harness is not to make the model omnipotent; it is to make model assistance usable inside an accountable procedure.

Cell 6: Round 1—independent diagnosis before solution generation

The first substantive round asks the agents to diagnose, not to legislate. Each agent receives targeted articles selected by specialty plus a rotating package designed to reduce attention concentration. The instruction is to identify concrete frictions, ambiguities, obsolete rules, proportionality problems, governance weaknesses or implementation problems and to link each issue to articles actually present in the authorized context.

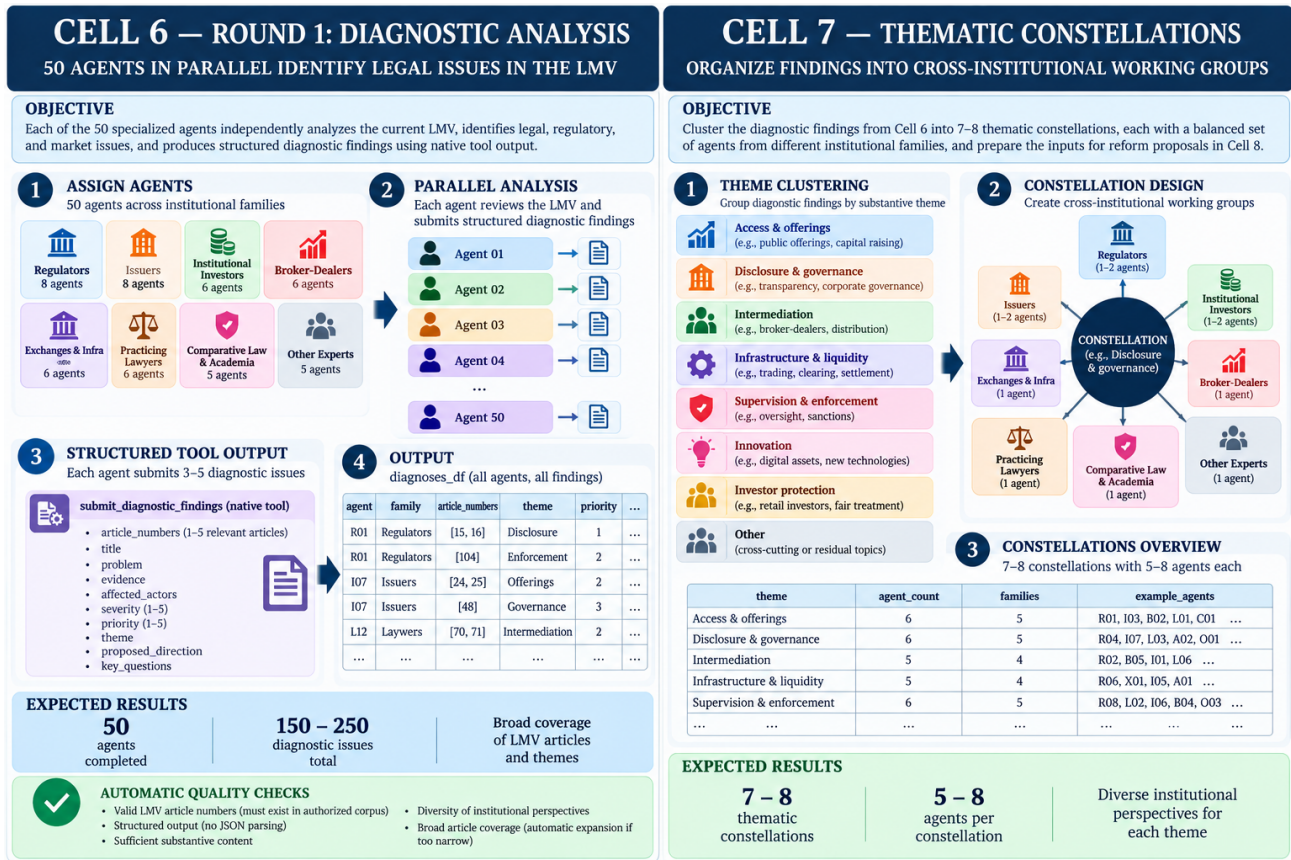


Figure 2: Cells 6–7: parallel independent diagnosis followed by a deterministic thematic synchronization barrier and formation of cross-institutional constellations.

This sequencing matters. If the model is asked for amendments too early, solution language can anchor the diagnosis. By forcing a diagnostic stage first, the notebook creates an intermediate evidence layer. A later reviewer can ask whether a proposal actually responds to a problem that multiple perspectives identified or whether it appeared only because the drafting model found the language attractive.

The stored run produced 176 usable diagnostic findings from 37 of the 50 configured agents and covered 169 distinct articles. The fact that only 37 agents contributed usable diagnoses is itself informative. Configured diversity is not the same thing as realized participation. A governance layer must therefore observe outputs, not merely count personas.

The diagnostic distribution reported by the outer loop was also uneven but not dominated by a single family:

Institutional family	Share of diagnostic findings
Regulators	17.0%
Issuers	16.5%
Broker-dealers	14.8%
Institutional investors	14.2%
Practicing lawyers	14.2%
Exchanges & infrastructure	13.6%
Legislative & policy	9.7%

The stored governance report lists seven families in the usable diagnostic output even though eight were configured. That is not something to conceal. It is exactly the sort of participation gap that a governed swarm should surface.

Comparative-law and academic expertise remains part of the architecture and later review logic, but the run demonstrates why “we created eight perspectives” cannot be treated as proof that eight perspectives materially influenced a particular execution.

Cell 6 structures diagnosis as an evidence-producing stage in its own right. Each assigned agent receives a bounded package of statutory material and a mandate to identify a concrete legal or operational problem. The expected record separates the description of the problem from the policy objective that might justify action. It also asks who may benefit, who may bear costs, what implementation risks arise and how urgent the matter appears. That separation discourages a common failure in generative policy work: presenting a favored solution as though it were the diagnosis itself. A finding should remain intelligible even if a later committee rejects the proposed remedy.

Article assignment combines specialty with rotation. Specialty helps direct technically relevant provisions to agents more likely to ask useful questions, while rotation reduces the chance that familiar flagship articles absorb all attention. Neither method guarantees comprehensive review. An agent may misunderstand a provision, and a rotating package may place material outside its strongest competence. The purpose is to create a broad first pass whose limitations remain observable. The article dictionary from Cell 3 constrains references, and the output schema from Cell 5 requires the agent to state its reasoning in a form that later cells can compare.

Independence in this round is procedural rather than cognitive. Agents do not first read one another’s findings. That avoids an immediate cascade in which early language anchors all later diagnoses. Because the same underlying model is used, complete independence is impossible; correlated priors remain. Still, separating requests prevents direct copying and gives the outer loop a clearer basis for measuring whether several roles independently surfaced related concerns. The synchronization step occurs only after the individual records have been stored.

The run’s 176 usable findings should therefore be interpreted as a candidate problem map, not as 176 established defects in the LMV. Usability means that the records passed the notebook’s structural and article-reference checks. It does not mean that a Mexican securities-law expert verified the interpretation, that empirical harm was demonstrated or that amendment is the correct response. Some findings may duplicate one another, reflect contested policy preferences or overlook relevant secondary rules. Their value is that they convert broad model impressions into addressable claims that can be grouped, challenged and traced.

The participation result is equally instructive. Thirteen configured agents did not contribute usable diagnoses. A weak presentation might continue to advertise fifty active analysts. The notebook instead preserves the difference between configuration and output. A human reviewer can ask whether failures clustered in a family, whether prompts were too demanding, whether validation rejected plausible material or whether the assignment strategy produced uneven workloads. Those questions are part of governance because silence can alter the thematic map as much as speech.

Cell 6 therefore creates the raw agenda while refusing to treat agenda formation as neutral. The number of mentions, the breadth of families and the distribution across articles later affect which issues receive proposal attention and selection credit. By storing the diagnostic genealogy, the notebook makes those upstream influences visible. If a final article ranks highly because several agents raised it, a reviewer can inspect the original claims rather than accepting the count at face value. Diagnosis becomes a documented layer of deliberation, not a disposable prompt on the way to drafting.

Cell 7: from diagnoses to constellations—temporary coalitions rather than fixed committees

After the first round, a deterministic synchronization barrier groups findings by topic, measures institutional breadth, counts article mentions and forms temporary constellations. No LLM is required for this coordination step. Python organizes the evidence the agents have already produced.

This is an important design choice. The swarm is adaptive in composition but not self-sovereign. It may reorganize who works with whom; it may not rewrite the rules that define authority. In institutional terms, the system permits flexible committees without permitting those committees to amend their own jurisdiction.

The constellation mechanism also improves the quality of proposal generation. A securities-law issue is rarely owned by one profession. A disclosure rule may affect issuers, investors, exchanges and supervisors simultaneously. A custody rule may implicate intermediaries, infrastructure and investor protection. Cross-family constellations force a proposal to encounter neighboring concerns before it becomes a legislative draft.

ADAPTATION WITHOUT SELF-AMENDMENT

The notebook distinguishes two forms of flexibility. **Operational adaptation** is allowed: form new working groups, retrieve different articles, ask for additional criticism. **Constitutional adaptation** is not: the swarm cannot remove the human gate, relax the source boundary, alter the twenty-article mandate or grant itself authority to file legislation.

That distinction generalizes well beyond law. Autonomous financial systems also need the ability to adapt portfolios, models or routing while keeping risk limits and escalation rights outside the component being optimized.

Cell 7 begins only after the independent diagnostic records have been collected. It acts as a synchronization barrier: no proposal-making group is formed from partial or still-changing output. The notebook groups findings by theme, counts references to each article and identifies how many institutional families contributed to a concern. These operations are performed with deterministic code so that the organization of the evidence can be reproduced and inspected. The language model may have created the diagnoses, but it does not decide invisibly which ones count as the same issue or which coalition receives priority.

The thematic categories provide an intermediate level between individual findings and statutory articles. Several provisions may participate in one policy problem, and one provision may attract concerns from different policy domains. The grouping mechanism therefore supports navigation rather than claiming a final legal taxonomy. A production implementation would likely combine rule-based labels, embeddings and human validation, especially where issues overlap. The pedagogical notebook's value lies in showing that coordination itself must be designed. Simply asking fifty agents to "collaborate" would obscure how their contributions were combined and whose framing governed the conversation.

Temporary constellations are assembled to cross institutional boundaries. A group focused on access and offerings should not consist only of issuer-oriented agents; a group focused on enforcement should not consist only of regulators. Bringing different families into the same proposal stage increases the chance that a candidate reform carries its principal trade-offs from the beginning. The constellation does not vote or acquire a corporate personality. It is an assignment structure that determines which diagnoses and perspectives are placed together for a bounded task.

The distinction between breadth and volume is important. Ten findings from one family may reveal intensity, while three findings from three families reveal cross-perspective reach. Cell 7 preserves both measures so later scoring can use them separately. Neither is decisive. Repetition may result from duplicated prompts, and institutional breadth may still reflect correlated model behavior. The measures are signals for review, not substitutes for judgment. Their transparency allows a human committee to alter weights or reject a theme whose apparent support is methodologically weak.

The synchronization barrier also creates a clean audit point. Before proposals exist, a reviewer can inspect the problem map and ask whether important families, articles or issue classes are missing. If the map is inadequate, the proper response is to rerun or supplement diagnosis, not to hope that drafting will repair the omission. This staged design contains error. It makes the transition from problem discovery to solution generation explicit and therefore contestable.

Most importantly, Cell 7 permits adaptation without self-amendment. The system can create new constellations in response to the evidence, balance assignments and request more attention for thinly covered topics. It cannot redefine the authorized corpus, change the human gate, reduce the required review separation or decide that twenty articles are no longer the mandate. Those higher-order rules remain fixed by the notebook operator. The cell thus models a familiar institutional arrangement: committees may organize their work flexibly, but their jurisdiction and decision rights come from an external charter.

By the end of Cell 7, the notebook has not endorsed a reform. It has produced a structured agenda and a set of temporary cross-family work groups. That restraint is methodologically useful. It ensures that later proposals can be linked to prior diagnoses and that collaboration begins from preserved evidence rather than an unrecorded conversational consensus. The constellation is a bridge between plural observation and focused design, not a shortcut from model output to legislative authority.

Cell 8: Round 2—reform proposals as structured hypotheses

In Round 2, cross-institutional constellations convert diagnoses into preliminary reforms. The output is still not legislative text in final form. It is a structured hypothesis about how a particular article might be changed and why.



Figure 3: Cell 8: parallel proposal generation, article-coverage expansion, deterministic proposal identity and a hard requirement to create a candidate universe larger than the final twenty-article mandate.

The proposal stage includes an expansion mechanism. If the thematic constellations do not produce a sufficiently broad candidate universe, the notebook identifies diagnosed articles not yet covered and requests additional article-specific proposals. The purpose is to avoid a selection process that merely ranks a thin and path-dependent first batch.

The stored run produced 33 candidate proposals for 33 distinct candidate articles. This is enough to create a real selection problem: the final twenty must be chosen from a larger feasible set. The architecture therefore separates *generation* from *portfolio construction*.

Each proposal preserves the article identifier under deterministic control. The model contributes the problem statement, policy objective, preliminary language, legal rationale, expected effects, implementation risks, dependencies, supporting roles, opposing roles and unresolved questions. This structured record becomes the object reviewed in the next round.

A key methodological point follows. The swarm does not create value merely because it produces more text. It creates value by transforming an initially broad legal corpus into progressively more structured intermediate objects:

Corpus → Diagnoses → Proposals → Reviews → Shortlist → Drafts.

Each transition has its own rules and its own evidence. That layered representation makes it possible to ask where an error entered the process.

Cell 8 converts the organized problem map into a candidate reform universe. The constellation receives the diagnoses relevant to its theme, the authorized article text and a schema that distinguishes the legal problem, policy objective, proposed operation, preliminary language, expected effects and unresolved dependencies. The task is generative, but it is not an invitation to write a complete bill. The proposal is an intermediate design object whose purpose is to be criticized and compared. Keeping it preliminary lowers the cost of disagreement: reviewers can reject an approach before polished legislative language creates false momentum.

Proposal identifiers are created outside the model and linked to a specific article. This prevents later prose from obscuring which legal object is under consideration. If two constellations propose different approaches to the same provision, they remain separate proposals until the article-level selection rule compares them. If one proposal mentions several related provisions, the notebook still requires a designated target and records other provisions as dependencies. That discipline is important because the final mandate concerns twenty distinct articles, not twenty narratives whose statutory scope is uncertain.

The expansion mechanism addresses path dependence. Initial constellations will naturally focus on themes with many diagnoses or clearer institutional narratives. Without a corrective, the final shortlist could be constrained by whatever happened to receive attention first. Cell 8 checks the size and article coverage of the candidate set. When the set is too narrow, it identifies diagnosed provisions that lack proposals and commissions additional article-specific work. Expansion does not guarantee a representative portfolio, but it ensures that the selection stage has alternatives and that the final twenty are not simply the first twenty proposals generated.

The stored set of 33 proposals for 33 candidate articles creates meaningful scarcity. Every candidate cannot advance, so the later scoring and human gate have real work to do. This is methodologically preferable to generating exactly twenty proposals and then describing all twenty as selected. A surplus exposes trade-offs, makes ranking consequential and provides a basis for sensitivity analysis. A future implementation could create several competing proposals per article, but the stored run emphasizes breadth across provisions.

The schema preserves both advocacy and friction. Supporting roles explain why the idea may attract institutional backing; opposing roles and unresolved questions prevent the record from becoming one-sided. Expected effects are predictions or hypotheses, not established facts, and implementation risks must remain visible. Comparative references receive the source-status discipline established earlier. The proposal can say that another jurisdiction offers a functional model, but it must not imply verification when the relevant source is absent from the authorized corpus.

Cell 8 also clarifies what the language model is good at. It can synthesize several diagnoses, articulate a theory of change and draft a preliminary mechanism in language that specialists can interrogate. It is not allowed to determine that the theory is correct, that the predicted effect will occur or that political costs are acceptable. Those questions move to adversarial review, human analysis and later impact assessment. The proposal stage thus uses generative capacity while resisting the tendency to confuse completeness of expression with completeness of evidence.

At the end of the cell, the notebook has a portfolio of reform hypotheses with stable identities and traceable parent diagnoses. This genealogy lets a reviewer move backward from a proposal to the concerns that motivated it and forward to the criticism, score and drafting decision it later receives. If a candidate has no persuasive diagnostic foundation, that weakness can be identified before selection. Cell 8 therefore broadens the option space while making every option more accountable than a free-standing model suggestion.

Cell 9: Round 3—adversarial review and reviewer independence

Every proposal is then assigned to an agent from a different institutional family. Reviewer identity is chosen deterministically with a preference for counter-perspectives and a load-balancing rule intended to diversify review assignments. The model does not get to change the article being reviewed.



Figure 4: Cell 9: independent adversarial review. Every candidate is scored across eight dimensions and must preserve its strongest objection even when the reviewer recommends keeping it.

The review rubric has eight dimensions, each scored from one to five: legal coherence, investor protection, proportionality, implementability, legislative clarity, evidence quality, market impact and regulatory risk. For regulatory risk, the scale is inverted so that a higher score means lower or more manageable risk. The notebook then computes a simple mean:

$$R_i = \frac{1}{8} \sum_{k=1}^8 r_{ik}.$$

Reviewers must also identify the strongest objection, possible unintended consequences, missing dependencies and the status of comparative-law claims. Comparative support can be marked VERIFIED, HYPOTHESIS or NONE; the model is not allowed to convert an analogy into evidence merely by stating it confidently.

The stored run completed all 33 adversarial reviews and verified that no proposal was reviewed by the same institutional family that led it. The recommendation distribution was striking: only two proposals were marked KEEP; thirty-one were marked REVISE; none were reported as dropped in the displayed summary.

A HEALTHY RESULT IS NOT UNIVERSAL APPROVAL

The fact that 31 of 33 proposals required revision is not a failure of the swarm. It demonstrates that the adversarial layer was not designed as a ceremonial sign-off. A governance architecture that always confirms the work of its own agents is not review; it is self-certification.

Cell 9 turns criticism into a mandatory and structured phase. The assignment algorithm first excludes the family that led the proposal and then favors a counter-perspective while balancing workload. This does not make the reviewer independent in the legal sense, because the same underlying model and system operator remain involved. It does, however, prevent the most direct form of self-review: the institutional lens that framed the proposal cannot also

supply its formal adversarial score. The deterministic assignment can be inspected and rerun, which is preferable to letting the proposal author choose a sympathetic reviewer.

The eight dimensions make evaluation multidimensional. A proposal can be legally coherent but costly to implement, protective of investors but disproportionate for smaller issuers, or clear in drafting while weak in empirical support. Averaging the scores yields a compact ranking input, yet the component scores remain available so that compensation can be challenged. A high market-impact score should not automatically erase a serious constitutional concern. In a production decision process, some criteria might operate as thresholds rather than as additive factors. The notebook's simple mean is pedagogical and transparent, not uniquely correct.

Narrative fields preserve information that a number cannot carry. The strongest objection requires the reviewer to state the best reason against the proposal rather than list superficial caveats. Unintended consequences direct attention to behavior the reform may induce outside its stated objective. Missing dependencies identify other statutes, regulations, institutions or technical systems that the proposed change assumes. These fields remain important even when the recommendation is **KEEP**; endorsement does not erase the duty to document residual risk.

The comparative-source status is another safeguard. A reviewer can recognize that a foreign analogy appears relevant while labeling it a hypothesis pending source verification. This prevents the scoring stage from laundering an unsupported claim into apparent evidence. The status also gives human researchers a concrete follow-up list. Instead of rereading every review to find uncertain references, they can filter the exported records for comparative claims that still require verification.

The recommendation vocabulary is intentionally limited. **KEEP** means that the proposal may proceed under the current record; **REVISE** means that the underlying issue may be worth pursuing but the design needs correction; **DROP** means that the candidate should not advance under the stated rationale. These labels organize workflow rather than announce legal truth. The stored distribution, with thirty-one revisions and only two keeps, shows that the reviewer prompt was capable of resisting automatic approval. It also foreshadows the later result in which selection can include proposals that still require work.

That last point is important. Cell 9 does not necessarily remove every **REVISE** candidate. The recommendation becomes one factor in a broader article-level score, and the strongest objections remain attached to the record. This design preserves potentially important agenda items even when their first formulation is inadequate. It also places a burden on later drafting: selection is permission to continue working on an article, not a declaration that the preliminary text survived review unchanged.

By completing all 33 reviews and verifying cross-family separation, the cell produces a fully paired candidate-and-critique dataset. Human readers can inspect disagreements rather than receiving only aggregate scores. The system's contribution is therefore not consensus but organized contestability. A proposal gains procedural credibility because it has an identified critic, a visible rubric and preserved objections, while ultimate substantive credibility still depends on expert human judgment and additional evidence.

Cell 10: the outer loop—governing the process rather than choosing the law

Cell 10 observes the swarm from outside the substantive debate. Its purpose is to detect process risk: concentration of participation, narrow institutional coverage, unsupported articles, duplicate proposals, unanswered objections and consensus cascades.

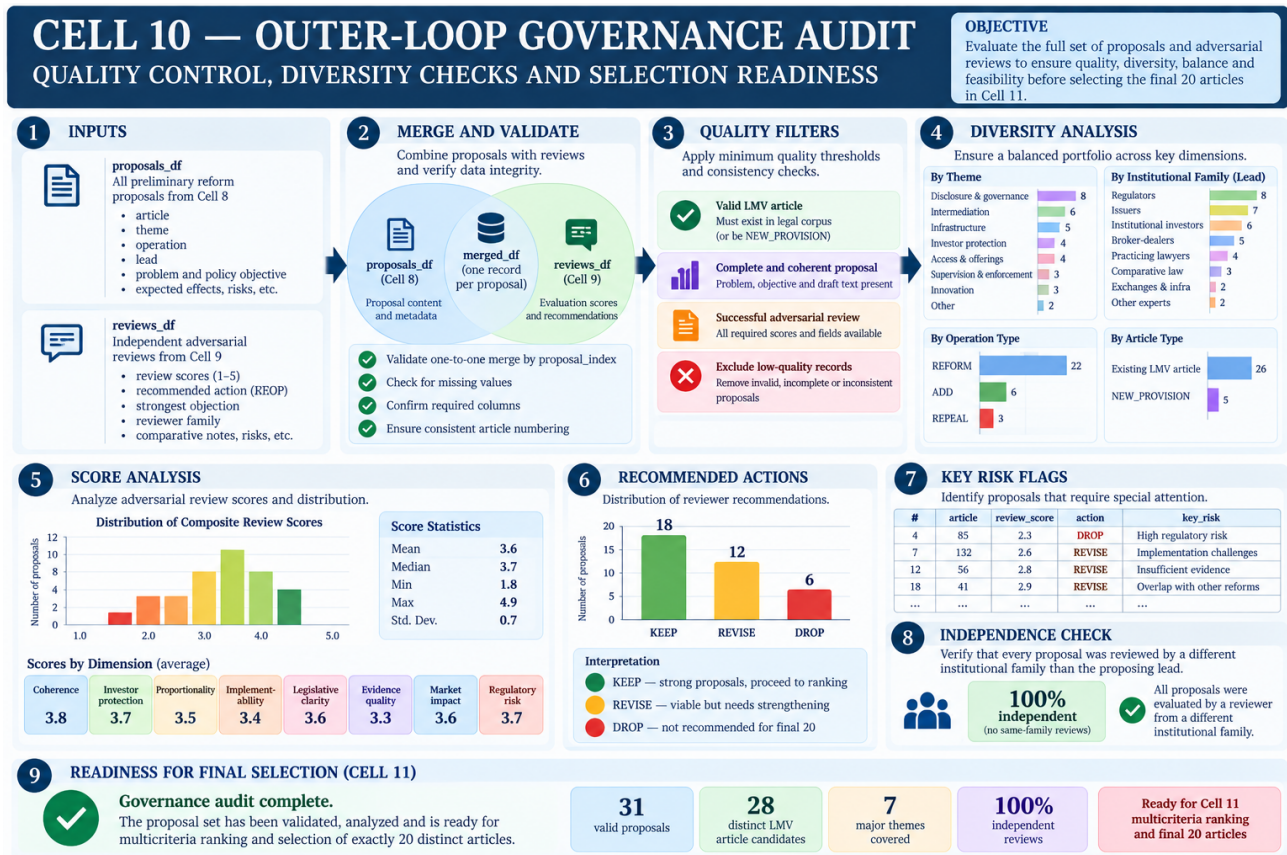


Figure 5: Cell 10: second-order governance. The outer loop evaluates the health of the deliberative process without acquiring authority to rewrite proposals.

The stored governance report recorded 176 diagnostic findings, 37 contributing agents, seven represented diagnostic families, 169 diagnosed articles, 33 proposals and 33 reviews, with no warnings. The largest family share of diagnostic issues was 17 percent. On those metrics, the outer-loop gate passed.

The conceptual importance of this layer is larger than the specific metrics. A multi-agent system introduces failure modes that do not exist in a single model call. Agents may herd. One perspective may dominate because it is more verbose. Multiple constellations may duplicate the same idea and create the illusion of independent support. A model may criticize its own earlier language in superficially different terms. A strong system therefore monitors *the distribution of influence* as well as the quality of individual outputs.

The outer loop is deliberately unable to alter the text of a proposal or promote a candidate to the final package. It can request more evidence, another critique or a different specialist. This preserves a separation between process supervision and substantive authorship.

SECOND-ORDER GOVERNANCE

In an ordinary workflow, the question is “Is this proposal good?” In an autonomous institutional workflow there is an additional question: “Is the process that produced and evaluated this proposal itself behaving well?” The outer loop exists to answer the second question.

Cell 10 aggregates operational evidence from the earlier rounds and asks whether the deliberative process met its own governance conditions. It counts contributing agents and represented families, measures the share of findings associated with each family, checks article support, searches for duplicate or near-duplicate proposals and confirms that review assignments respected separation rules. These checks do not evaluate the wisdom of a reform. They evaluate whether the conditions under which reforms were generated are sufficiently diverse, traceable and complete

to justify moving forward.

Participation metrics are essential because nominal architecture can diverge from actual execution. Cell 4 created fifty agents in eight families, but Cell 6 produced usable diagnoses from only 37 agents and seven families. The outer loop makes that discrepancy visible. A threshold can trigger a warning when one family dominates, when too few agents contribute or when a configured perspective disappears. The stored run did not trigger the implemented warnings, but the report still preserves the raw counts so that a human reviewer can apply a stricter standard.

Concentration analysis should be interpreted carefully. A seventeen-percent family share is not inherently improper, and equal shares would not prove fairness. Some statutory themes may legitimately attract more attention from particular specialists. The metric is a prompt for examination, not a constitutional formula. Its value lies in detecting extreme or unexplained concentration that would otherwise be hidden by the total volume of output. The same caution applies to article frequency: repeated mentions can signal importance, duplication or prompt-induced herding.

Duplicate detection addresses a distinctive multi-agent risk. Several groups may express substantially the same idea with different wording, creating an illusion of independent alternatives or broad support. By flagging overlap, the outer loop can request consolidation or additional differentiation before scoring. A production system would need robust semantic methods and human confirmation because legal proposals can appear similar while differing in scope, exceptions or institutional allocation. Even a simple duplication check, however, demonstrates that output volume should not be mistaken for deliberative diversity.

The audit also looks for unsupported or unanswered elements. A proposal tied to an article absent from the corpus, a review that omits its strongest objection or a constellation without cross-family participation may indicate a procedural defect. The outer loop can stop progression, request another review or send an issue back to an earlier stage. It cannot repair substantive text on its own. This restriction preserves the independence of supervision: the monitor identifies conditions and invokes a remedy defined by the workflow rather than becoming another untracked author.

Consensus-cascade risk provides the broader rationale. When agents share a model and later stages consume earlier outputs, the system can amplify an initial framing through repetition. Cell 10 cannot measure true cognitive independence, but it can examine proxies such as family breadth, assignment separation, duplicated language and concentration. Those signals help a human reviewer distinguish a robustly contested issue from an idea that became prominent because the architecture kept recirculating it.

The outer-loop report is therefore a governance artifact, not a certificate of legitimacy. Passing means that the implemented procedural checks found no blocking defect. It does not mean that all relevant interests were represented, that diagnoses were correct or that the process would satisfy public-law consultation requirements. Cell 10's disciplined modesty is part of its design. It observes the swarm, records process health and enforces stop conditions while leaving the choice of law outside its mandate.

Cell 11: multicriteria selection—from 33 candidates to exactly 20 articles

Cell 11 turns reviewed proposals into a portfolio. The mandate is not to select twenty pieces of text; it is to select exactly twenty *distinct existing LMV articles*. For each article, the notebook keeps the highest-scoring proposal and then ranks the resulting article-level candidates.

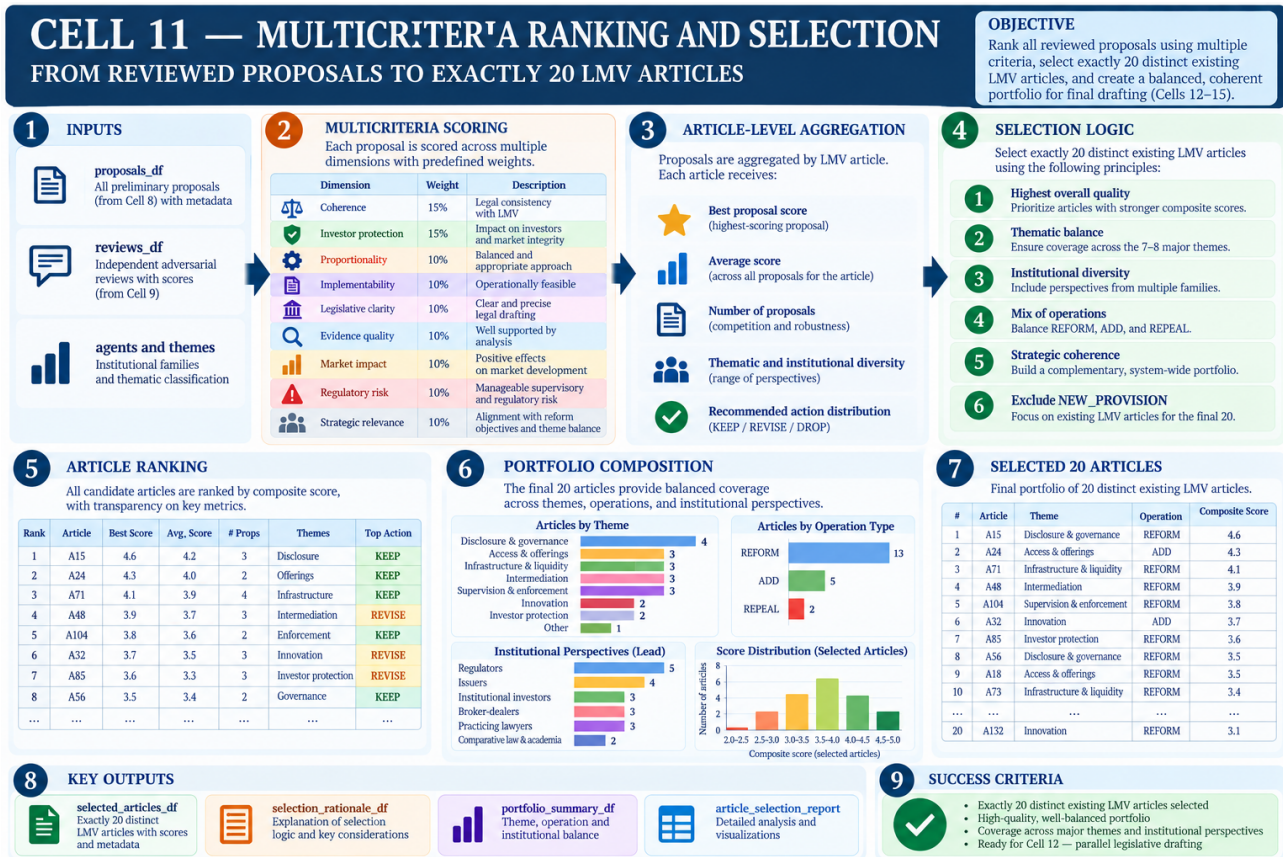


Figure 6: Cell 11: article-level aggregation, multicriteria ranking and the explicit human authorization gate before legislative drafting.

The selection score combines three forms of information: review quality, breadth of diagnostic support across institutional families, and frequency of diagnostic mentions. It also adds an adjustment based on the adversarial recommendation:

$$S_i = 0.70R_i + 0.75\frac{F_i}{F_{\max}} + 0.75\frac{M_i}{M_{\max}} + A_i,$$

where R_i is the eight-dimension review score, F_i is the number of supporting diagnostic families, M_i is the diagnostic-mention count, and

$$A_i = \begin{cases} +0.30 & \text{if KEEP,} \\ 0 & \text{if REVISE,} \\ -0.75 & \text{if DROP.} \end{cases}$$

The weighting intentionally prevents raw popularity from overwhelming review quality. Diagnostic breadth and mentions matter, but each is normalized. The score is visible and contestable rather than presented as an opaque “AI ranking.”

The selected portfolio in the stored run contained the following twenty provisions.

Article	Theme	Selection	Review	Action
29	Access & offerings	3.6625	3.625	REVISE
71	Access & offerings	3.4750	3.250	REVISE
190 Bis	Supervision & enforcement	3.4000	3.250	KEEP
217	Supervision & enforcement	3.3375	3.375	KEEP
126	Supervision & enforcement	3.3000	3.750	REVISE
96	Supervision & enforcement	3.1000	3.250	REVISE

Article	Theme	Selection	Review	Action
12	Other	3.1000	3.250	REVISE
4	Other	3.0500	2.750	REVISE
173 Bis	Supervision & enforcement	3.0375	3.375	REVISE
123	Infrastructure & liquidity	3.0375	3.375	REVISE
66	Supervision & enforcement	3.0375	3.375	REVISE
144	Other	3.0125	3.125	REVISE
305	Supervision & enforcement	3.0125	3.125	REVISE
108 Bis	Supervision & enforcement	2.9500	3.250	REVISE
275	Supervision & enforcement	2.8750	2.500	REVISE
201	Access & offerings	2.7750	3.000	REVISE
64 Bis 2	Access & offerings	2.7750	3.000	REVISE
160	Access & offerings	2.7750	3.000	REVISE
177	Supervision & enforcement	2.6000	2.750	REVISE
180	Infrastructure & liquidity	2.6000	2.750	REVISE

The notebook then places an explicit human gate between selection and drafting. This is more than a user-interface pause. It encodes a constitutional principle for the system: quantitative ranking can organize judgment without becoming authority. The swarm may say which twenty articles rank highest under the declared method. It cannot convert that ranking into a legislative mandate on its own.

Cell 11 begins by changing the unit of decision from proposal to article. Where several proposals target the same provision, only the highest-scoring candidate for that article can represent it in the final ranking. This prevents one heavily discussed provision from occupying several of the twenty available places. It also aligns the computation with the stated mandate: exactly twenty distinct existing LMV articles. The article dictionary again supplies deterministic identity, so spelling variants or model-generated labels cannot inflate the count.

The score combines review quality with two signals inherited from diagnosis. The review mean receives the greatest weight because the proposal's legal and policy characteristics should matter more than raw attention. Family breadth rewards concerns that appeared across different institutional lenses, while mention frequency recognizes recurrence. Normalization keeps large counts within a bounded contribution. The recommendation adjustment gives a modest advantage to **KEEP** and a penalty to **DROP** without allowing the categorical label to replace the underlying rubric.

Every term in this formula expresses a normative choice. Seventy percent for review quality, fifteen percent for breadth and fifteen percent for mentions is not discovered by the model or dictated by securities law. It is a governance rule selected by the notebook designer. Transparency makes that choice contestable. A legislative body might prioritize constitutional feasibility, distributional impact or enforcement capacity through thresholds that cannot be offset by popularity. A researcher can rerun the ranking under alternative weights and measure which articles remain stable. The notebook supplies a baseline method, not an objective measure of legislative merit.

The displayed shortlist preserves both the composite selection score and the underlying review score. It also retains the recommendation, making clear that eighteen of the selected articles were associated with **REVISE** rather than **KEEP**. Selection therefore means that an article deserves drafting attention within the experiment, not that its current proposal has been approved. This interpretation is necessary to reconcile the portfolio with the adversarial record. The selection stage ranks agenda items under scarcity; it does not erase objections or transform weak preliminary language into acceptable law.

Exactly twenty is enforced as a hard constraint. If fewer than twenty distinct valid articles were available, the proper result would be a stop or a request for additional proposals, not duplication or invention. If more than twenty qualify, the score establishes an ordered candidate set and a cutoff. Ties and close margins should be visible because small weight changes may alter the last positions. A production process could reserve places by theme or require minimum representation across policy domains, but any such rule should be declared before results are known.

The human gate is the decisive feature. After the ranking is displayed, a human must authorize progression to legislative drafting. That reviewer can reject the list, request sensitivity analysis, exclude an article for reasons outside the notebook's evidence or demand further diagnosis. If the reviewer changes the portfolio, the change should be recorded with reasons rather than attributed to the algorithm. Human control is meaningful only when it can interrupt or modify the path and when its intervention leaves an audit trail.

Cell 11 thus turns a large body of synthetic analysis into a bounded decision aid. It makes trade-offs visible, applies the article constraint consistently and produces a shortlist that can be inspected before expensive drafting begins. Its authority ends at recommendation. The formula organizes information; it does not enact a policy preference, certify legality or substitute for the accountable institution that must decide whether any of the twenty provisions should actually be amended.

Cell 12: legislative drafting—generation is deliberately a late-stage activity

Only after selection does the system produce article-level legislative language. Each authorized article is assigned to a drafter and to a reviewer from a different family. The drafter receives the current article text, the approved operation, the diagnosed problem, the policy objective, the preliminary Round 2 language and the adversarial critique.

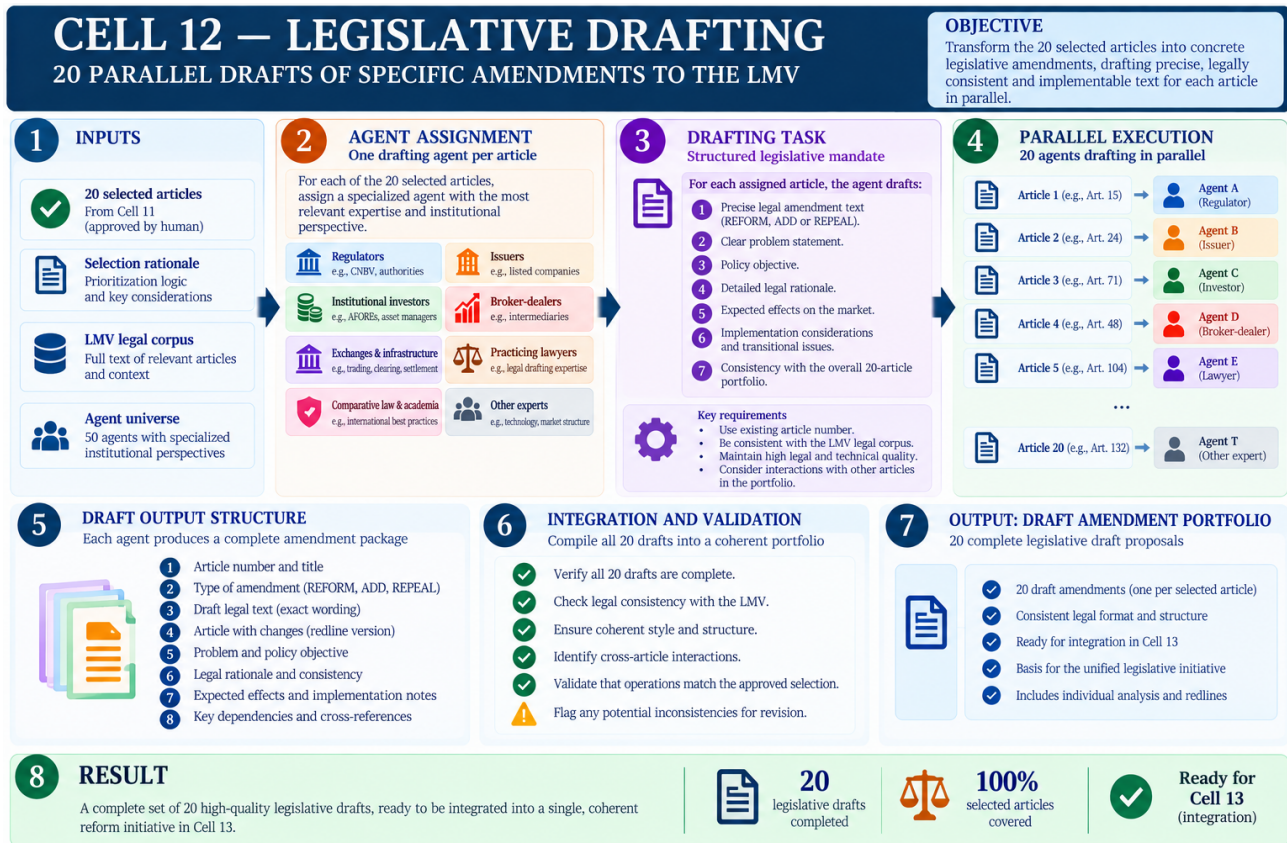


Figure 7: Cell 12: twenty parallel article-specific drafting tasks, each followed by an independent legislative-technique review and preservation of unresolved issues.

The drafting prompt imposes several constraints. Unrelated current text should be preserved. No other LMV article may be silently altered. Cross-references must be identified. Dependencies that might require changes elsewhere are recorded under `follow_up_required` rather than being inserted automatically. The status of the output remains an *anteproyecto* for human legal review.

The drafting run successfully produced twenty distinct article drafts. The notebook also reports that all twenty technique reviews were structurally complete. But substantive approval was much more cautious: one article was marked approved and nineteen were not. Article 173 Bis was the only draft shown with `technique_approved = True`; the remaining selected articles were flagged for revision.

This distinction is a major result of the experiment. “The system finished” and “the legal text is acceptable” are not synonyms. A robust workflow can preserve a valid draft even when the review layer rejects or criticizes it. The notebook intentionally does not delete flagged drafts because doing so would erase useful evidence about what the model tried to do and why reviewers objected.

THE MOST IMPORTANT NEGATIVE RESULT

The drafting stage did **not** produce twenty technique-approved amendments. It produced twenty drafts and nineteen unresolved technique-review flags. The architecture is valuable precisely because it makes that failure state visible instead of collapsing it into a polished final document.

Cell 12 treats legislative text as the output of accumulated constraints rather than the starting point of the exercise. Each drafter receives the authorized current provision, the selected proposal, its diagnostic genealogy and the adversarial review. This context forces the drafting task to respond to an identified problem while confronting the strongest recorded objection. It also reduces the temptation to solve unrelated issues encountered in the surrounding

statute. The instruction is narrow: draft the authorized operation for the selected article and disclose anything that cannot be completed within that scope.

Drafter assignment and technique-review assignment remain separate. The reviewer belongs to a different institutional family, which creates a second cross-perspective check after proposal review. The technique reviewer considers whether the amendment is clear, internally coherent, faithful to the approved objective and appropriately bounded. This review is not a substitute for constitutional analysis or official legislative counsel. It is an additional synthetic control designed to detect obvious defects before the material reaches humans.

Preservation of unrelated current text is a crucial constraint. Generative models often improve prose broadly, harmonize terminology or add apparently helpful provisions beyond the requested change. In legislation, those behaviors can create unauthorized legal effects. Cell 12 instructs the drafter to alter only what the selected operation requires. When a necessary change appears to involve another article, the model must identify it as a dependency under **follow_up_required**. The notebook thereby prefers an explicit incomplete dependency to a silent expansion of scope.

Cross-references receive similar treatment. A draft may rely on definitions, procedures or powers located elsewhere in the LMV or in secondary regulation. The model can flag those connections, but it cannot presume that every reference is current, complete or legally compatible. Human counsel must verify them against the full corpus and related law. By making cross-references a structured output, the cell creates a research agenda for that verification and prevents dependencies from disappearing inside explanatory prose.

The stored outcome sharply illustrates the difference between production and approval. All twenty selected articles received a draft, and all twenty technique-review records were structurally complete. Yet only Article 173 Bis received a positive technique flag. Nineteen drafts remained unapproved. This is not merely a statistic about model quality; it is evidence that the workflow permits an adverse control to survive. The notebook does not relabel a rejected draft as approved simply because a complete package is desired.

Nor does it delete the nineteen flagged drafts. Retention preserves the work product and the objections for human diagnosis. A legal team can compare the proposed text with the technique review, decide whether the defect is reparable and trace the issue back to the proposal or even the original diagnosis. Deletion would hide failure and force the team to reconstruct what happened. Keeping the draft does not endorse it; it makes the failure auditable.

The *anteproyecto* label governs the status of every output. Even a positively reviewed draft remains preliminary because the notebook has not completed constitutional review, regulatory-impact analysis, coordination with other statutes or political consultation. The cell therefore refuses a common category error: equating legislative style with legislative validity. A provision can sound formal and still be unlawful, incoherent or undesirable.

Cell 12's deeper lesson is that generation should occur only after agenda formation, criticism, selection and authorization. This ordering saves human attention and reduces the persuasive force of premature polished language. The resulting drafts are concrete enough to test but remain attached to their limitations. The model contributes linguistic capability inside a bounded assignment; technique review supplies structured resistance; deterministic code preserves scope and counts; humans retain authority to revise, reject or abandon every amendment.

Cell 13: integration—a coherent initiative without expanding the mandate

Cell 13 assigns an integrator from the legislative and policy family to assemble the twenty authorized article drafts into a title and an *exposición de motivos*. The integrator is explicitly prohibited from introducing, removing or renumbering articles.

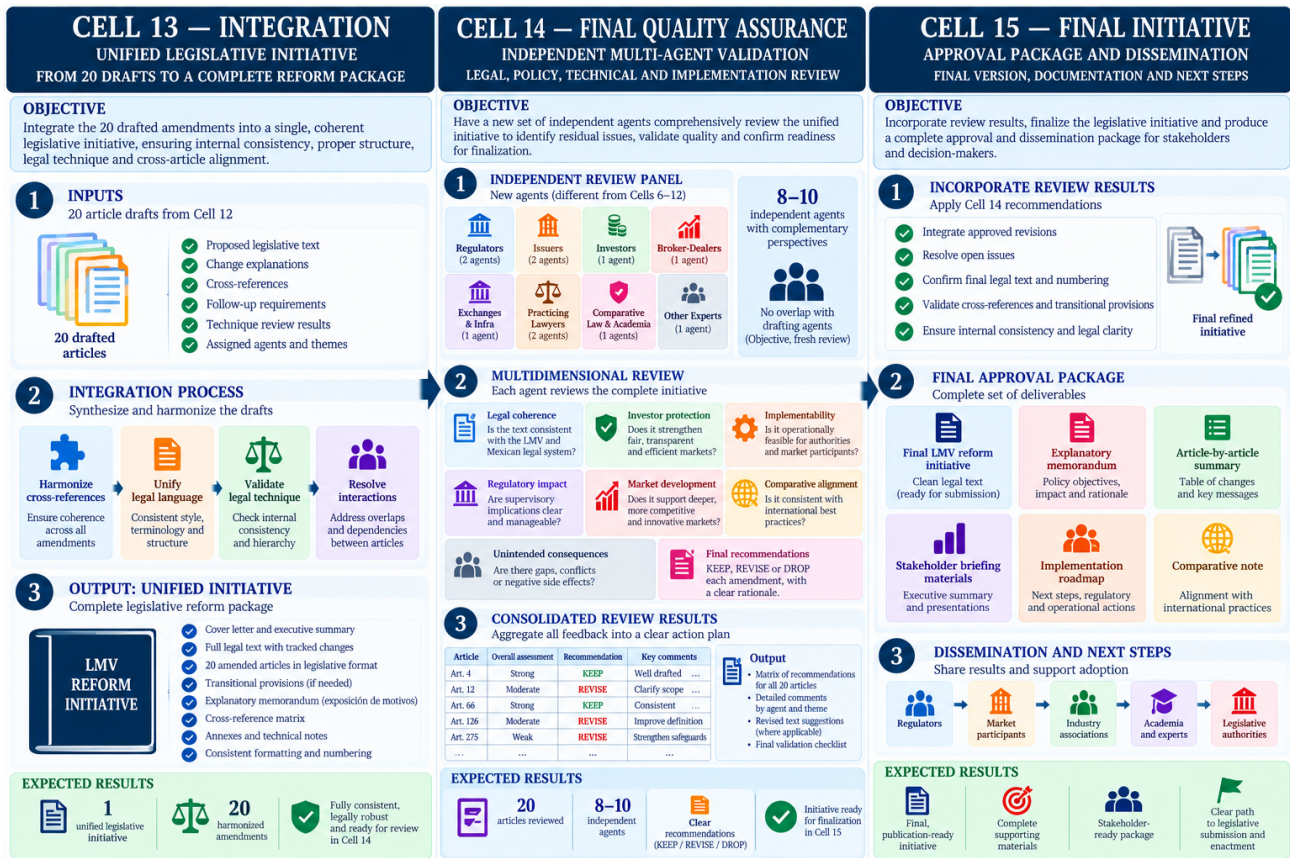


Figure 8: Cells 13–15: integration of the twenty-article package, deterministic quality assurance, and export of both the product and its audit trail.

The generated title in the stored run was a preliminary comprehensive reform to the LMV organized around legal clarity, supervisory proportionality and investor protection. The statement of reasons grouped the package around several recurring concerns: indeterminate legal terms, procedural gaps, delegation without sufficiently explicit statutory floors, cross-border responsibility, supervisory discretion and unequal compliance burdens.

The integrator’s job is therefore not merely editorial. A coherent statement of reasons is a test of whether the selected articles can be explained as one reform architecture rather than a collection of high-scoring but unrelated interventions. If the system cannot articulate a common theory of change, the portfolio may be numerically optimized yet institutionally incoherent.

The notebook also requires the integration output to acknowledge trade-offs. This matters because a credible reform package should not present every intervention as a Pareto improvement. Stronger disclosure can impose cost. More procedural rights can slow intervention. Proportionality may reduce burdens while complicating supervisory segmentation. The system is asked to preserve these tensions instead of writing advocacy copy for its own work.

Cell 13 receives the twenty article-level draft records only after the human gate and drafting-stage reviews. Its task is to assemble a navigable package: a provisional title, an organizing account of the reform and the authorized article texts. Integration is needed because independent drafting can produce inconsistent terminology, duplicated rationales or a collection of measures that lack a common theory. The integrator must surface those tensions without gaining permission to solve them by changing the portfolio.

The prohibition on adding, removing or renumbering articles is therefore central. A fluent integrator may be tempted to repair a gap by inserting a definition, dropping an awkward draft or expanding a cross-reference. Those actions would bypass Cell 11’s selection rule and the human authorization attached to it. Cell 13 can identify the need for further work, but substantive changes must return through the governed process. Deterministic comparison of the integrated article set against the selected set provides a mechanical check on this boundary.

The *exposición de motivos* performs more than a presentational function. It states the problems the package claims to address, the principles that connect the amendments and the expected institutional effects. Because the twenty articles were selected through individual scores, the narrative tests whether they can plausibly be defended as one reform program. If the account becomes vague or contradictory, that may reveal that the portfolio is an assortment rather than a coherent initiative. Integration thus acts as a diagnostic of selection as well as an editorial stage.

At the same time, the statement of reasons must not convert hypotheses into facts. Expected benefits remain contingent, and unresolved objections must not disappear because the package now has a title. The integrator is asked to acknowledge costs, distributional effects and implementation burdens. It should distinguish legal clarification from empirical prediction and note where impact evidence is missing. This discipline is especially important because integrated prose can create an impression of confidence stronger than any individual record supports.

Terminological consistency is another focus. Separate drafters may use different expressions for similar actors, procedures or standards. The integrator can flag those inconsistencies and harmonize purely editorial elements when authorized, but a change that affects legal meaning requires review. In a production setting, a version-controlled redline and legislative-counsel review would be necessary. The notebook demonstrates the need for that function without claiming to automate it conclusively.

The technique flags from Cell 12 remain attached to the package. Integration does not reset them. Nineteen unresolved reviews therefore continue to constrain how the assembled document is described. The correct status is a preliminary instrument for human examination, not a filing-ready initiative. Preserving that status across stages prevents a downstream polished artifact from laundering upstream defects.

Cell 13 ultimately creates a coherent container without granting the container new authority. It makes the selected work easier to read, reveals common themes and exposes inconsistencies that only become visible at portfolio scale. Yet it remains subordinate to the approved article list, the source corpus and the review record. The integrator can explain the package and organize it; it cannot legislate around the controls that produced it.

Cell 14: validation—deterministic integrity is not a legal opinion

Cell 14 performs a final integrity gate. It checks that there are exactly twenty selected articles, exactly twenty distinct drafts, that the selected and drafted sets match, that selected articles exist in the authorized corpus, and that every draft contains both proposed text and an explanation of the change.

The resulting report in the stored run contained no *critical* deterministic errors. It therefore set `ready_for_human_review` to true. At the same time, the findings section explicitly preserved the nineteen article-level technique flags and added five mandatory areas of further work:

- constitutional review;
- consistency with secondary regulation and other Mexican statutes;
- regulatory-impact analysis;
- source-level verification of comparative-law claims;
- consultation with political actors, market participants and authorities.

The meaning of the status flag must therefore be precise. It means that the package is mechanically coherent enough to be handed to a human legal team. It does not mean that the proposals are legally correct, constitutionally sound, politically desirable, economically efficient or technically ready for filing.

A WELL-DESIGNED STOPPING RULE

The system is allowed to finish by saying: “Here is a traceable package that passed structural checks and still contains unresolved legal issues.” That is a stronger institutional behavior than forcing every run to end with a confident recommendation.

The source notebook also warns about hallucination signals: article citations that cannot be recovered, references to authorities absent from the corpus, unsupported comparative claims and overly categorical language. No automated test can certify legal validity. But mechanical checks can prevent errors that should never survive to human review.

Cell 14 is a final structural gate applied after integration. It compares the selected, drafted and integrated article sets; verifies distinctness and the exact count of twenty; checks every identifier against the corpus; and confirms that required fields such as proposed text and explanation are present. These are crisp conditions with unambiguous machine-level answers. Assigning them to deterministic code avoids the circularity of asking the model whether it followed its own instructions.

Set equality is particularly important. Twenty drafts can exist while still failing the mandate if one selected article is missing and another appears twice. An integrated document can look complete while silently omitting a difficult provision. Cell 14 tests membership rather than appearance. It can therefore detect structural substitutions that a human reader might overlook in a long package. The same logic confirms that the integrator did not introduce an unselected provision.

Required-field checks establish minimum completeness, not substantive adequacy. A nonempty explanation may still be unpersuasive, and a block of proposed text may still be legally defective. The validation report should consequently name each test precisely. “Draft present” is a different claim from “draft valid.” The notebook’s status language reflects this distinction by declaring readiness for human review rather than readiness for enactment or filing.

The nineteen negative technique flags remain findings even though they are not treated as deterministic critical errors. This design allows the package to reach human reviewers with its shortcomings intact. Whether that is the appropriate stopping rule depends on institutional purpose. A production legislature might require all technique reviews to pass before integration; a research laboratory may deliberately export failed drafts so experts can study them. The key is that the rule is explicit and that `ready_for_human_review = true` cannot be misquoted as substantive approval.

The five mandatory areas of further work define the limits of automated validation. Constitutional compatibility requires specialized legal analysis and may depend on jurisprudence outside the immediate corpus. Consistency with other statutes and secondary rules requires a broader source set. Regulatory impact demands empirical assumptions, data and distributional analysis. Comparative claims require source-level verification. Consultation requires real institutions and affected people, not synthetic stand-ins. Cell 14 records these needs because structural success cannot resolve them.

Hallucination signals provide another layer of caution. An unrecoverable citation, absent authority or categorical statement unsupported by context should trigger investigation. Automated routines can flag patterns, but they

cannot reliably decide whether a legal inference is sound. A citation may be recoverable and still be misinterpreted. Validation therefore narrows avoidable error without promising semantic certainty.

The report also supports reproducibility. A later rerun can compare which checks passed, which findings persisted and whether repairs changed the status. Each remediation should update the relevant artifact rather than merely alter the final prose. This creates a review loop in which defects have identifiable locations and resolutions.

Cell 14's institutional contribution is an honest stopping rule. It permits the system to say that a package is structurally complete enough to inspect while still legally unfinished. That statement is less dramatic than automated certification, but it is more useful and defensible. Deterministic validation protects the handoff to humans; it does not displace the judgment those humans must exercise.

Cell 15: export and traceability—the product is an expediente, not only a document

The final cell exports ten files: the Round 1 diagnoses, Round 2 proposals, Round 3 adversarial reviews, the twenty-article shortlist, the twenty article drafts, the audit log, the governance report, the validation report, a structured preliminary bill and a human-readable Markdown *anteproyecto*. The notebook then copies the output directory to Google Drive.

This export design is methodologically significant. A generated legal document by itself is difficult to audit. The exported bundle instead preserves a genealogy. A reviewer can ask:

- Which diagnostic issue first raised Article 217?
- Which families supported or objected to the proposal?
- What did the adversarial reviewer identify as the strongest objection?
- Why did the article rank inside the top twenty?
- Who drafted it and who reviewed the drafting technique?
- Which dependencies were left unresolved?

The answer should be recoverable from artifacts rather than from model memory. This is the shift from conversational AI toward institutional AI: the system produces records that support governance after the model call has ended.

TRACEABILITY AS A FIRST-CLASS OUTPUT

The final initiative is only one output of the laboratory. The audit trail, evidence map, objections, selection logic and validation findings are equally important because they make disagreement reconstructable. A third party can challenge a decision without having to trust a black box.

Cell 15 converts the notebook’s in-memory state into a durable review bundle. The ten exported files separate major artifact classes instead of collapsing everything into one polished document. Diagnoses preserve the initial problem map. Proposals preserve theories of change. Reviews preserve objections and scores. The shortlist records the selection decision. Draft files record article-level language and technique findings. Governance, audit and validation reports document how the process behaved. Structured and human-readable versions of the preliminary initiative support both computation and legal reading.

This separation makes the bundle usable by different reviewers. A legislative drafter may focus on article text and cross-references. A governance reviewer may examine participation, assignment and prompt hashes. An economist may inspect expected effects and identify assumptions requiring impact analysis. A project manager may use the validation findings as a remediation list. No reviewer is forced to infer the entire genealogy from the final narrative, and no single document is asked to serve every evidentiary function.

Stable identifiers connect the files. An article or proposal should be traceable across diagnosis, review, scoring, drafting and integration. This relational structure is what turns a collection of exports into an *expediente*. If an identifier changes silently or a file loses its parent reference, the chain becomes harder to audit. A production implementation would add a manifest containing file hashes, schema versions, creation times and the corpus hash, allowing a recipient to verify that the bundle is complete and unaltered.

Copying the output directory to persistent storage protects the record from the temporary lifecycle of a Colab runtime. Persistence, however, creates responsibilities. Credentials and unnecessary prompt content should not be exported. Access controls, retention periods and personal-data review would be necessary if real submissions or confidential materials were used. The pedagogical notebook operates with synthetic roles, but its export architecture anticipates the governance questions that arise when records leave the execution environment.

Machine-readable formats are important for reanalysis. A reviewer can recalculate rankings under different weights, filter unverified comparative claims or count unresolved dependencies without extracting data from prose. The Markdown initiative serves the complementary purpose of human accessibility. Keeping both forms reduces the risk that presentation and evidence diverge. If a number in the narrative differs from the structured file, the discrepancy can be investigated rather than hidden.

The audit trail also supports accountability after model calls have ended. Language-model memory is not the record of the institution, and a chat transcript alone would not reliably show which version of a proposal reached selection. Exported artifacts freeze the state of each stage. They make it possible to reconstruct why Article 217 advanced, which reviewer objected, what score was used and whether the final draft addressed the objection. That reconstructability is more valuable than a generic claim that the system was “explainable.”

Export does not solve every provenance problem. A file can be complete while its underlying legal analysis remains wrong. An operator can also choose not to distribute inconvenient artifacts. Institutional deployment therefore requires policies specifying which files must accompany any public or internal presentation of the initiative. The notebook models that norm by exporting product and process together. The final bill is not treated as separable from its limitations.

Cell 15 completes the pipeline without converting completion into approval. Its output is a package ready to be examined, challenged, corrected and, if necessary, rejected by humans. The durable record allows disagreement to continue outside the notebook and enables later reruns to be compared with the stored execution. In that sense, export is the last governance act: the swarm relinquishes control of its artifacts while preserving enough context for accountable institutions to decide what, if anything, should happen next.

What the experiment teaches about autonomous systems

Although the domain is legal reform, the architecture is an autonomous-system architecture. It combines world representation, specialized perception, proposal generation, critique, ranking, bounded execution, monitoring and human override. The legal case makes the authority problem unusually clear because no one would seriously claim that statistical ranking should itself acquire legislative power.

Several broader lessons follow.

Capability and authority must be designed separately

A model may be capable of drafting an amendment while lacking authority to decide that the amendment should exist. This sounds obvious in law, yet many autonomous systems blur the distinction. In finance, a model that can predict returns is not thereby authorized to exceed a risk limit. In operations, a model that can optimize a route is not thereby authorized to ignore safety constraints. The notebook makes the separation executable.

Multiple agents create new governance problems

A swarm can broaden analysis, but it also creates coordination risk. Correlated model error can masquerade as consensus. Verbosity can create influence. Repeated variants can inflate apparent support. Reviewers can become overloaded. Temporary coalitions can lock into early framings. This is why the outer loop is not optional.

The most valuable automation may be the structure around the model

The notebook's strongest design elements are not exotic model behaviors. They are mundane institutional controls: an authorized corpus, unique identifiers, structured schemas, reviewer separation, visible scores, hard article counts, audit logs and stop conditions. These mechanisms constrain a general model into a task-specific system.

Good autonomous systems preserve negative information

The nineteen technique-review flags are not noise to be cleaned away. They are part of the result. The architecture becomes more trustworthy when it preserves reasons not to proceed.

Interpreting the twenty-article portfolio

The twenty selected articles should be understood as the output of one bounded experimental run, not as an independently validated statement about the current LMV. Their importance for this monograph is methodological: they demonstrate how the pipeline converts a broad statute into a finite portfolio under visible scoring rules.

The portfolio was concentrated in two themes. Supervision and enforcement accounted for the largest group, including Articles 190 Bis, 217, 126, 96, 173 Bis, 66, 305, 108 Bis, 275 and 177. Access and offerings contributed Articles 29, 71, 201, 64 Bis 2 and 160. Infrastructure and liquidity contributed Articles 123 and 180. Articles 12, 4 and 144 were classified under the residual “Other” category.

That distribution is itself a research object. It can reflect true concentration of frictions, but it can also reflect corpus segmentation, prompt design, the composition of the agent families, retrieval terms or the salience of certain regulatory concepts to the underlying model. A future research design should therefore compare portfolios across repeated runs, different random seeds, different model families and different agent compositions.

DO NOT CONFUSE SELECTION WITH VERIFICATION

This monograph reports the notebook’s selected article numbers and scores because they are part of the experiment. It does not independently verify that the saved run’s proposed legal amendments are correctly mapped to the current Mexican statute. That work belongs to the subsequent human legal review explicitly required by the notebook itself.

The fact that the later legal-drafting review flagged nineteen of twenty provisions reinforces this point. Ranking is useful for allocating attention. It is not a substitute for doctrinal verification.

Limitations: what fifty synthetic perspectives cannot provide

The notebook is explicit that it is a pedagogical and research laboratory. Several limitations should remain visible.

First, the fifty agents are not fifty humans. They do not possess independent professional histories, private information, political mandates or real economic exposure. They are structured simulations of perspectives generated by one model. The architecture can organize disagreement; it cannot manufacture democratic representation.

Second, the primary unit of legal memory is the article. Production legal analysis would require more granular parsing of paragraphs, fractions, subparagraphs, transitory provisions and cross-references. It would also require version control across amendments and systematic reconciliation with secondary regulation and other statutes.

Third, a single model family creates correlated-error risk. Different personas can still converge on the same latent assumptions. True model diversity, human expert review and empirical market evidence would strengthen the design.

Fourth, the scoring function is normative. The weights—70 percent adversarial-review quality, 15 percent normalized family breadth and 15 percent normalized diagnostic mentions, plus the recommendation adjustment—are transparent but not uniquely correct. A different institution might reasonably choose different weights or a non-compensatory decision rule.

Fifth, the pipeline does not perform a regulatory-impact analysis. It does not estimate compliance cost, market liquidity, litigation risk, competitive effects or transition burden with empirical data. Those analyses should become separate tools rather than being inferred from persuasive prose.

Sixth, political feasibility is outside the system. Legislative reform is not a technocratic optimization problem alone. Committee jurisdiction, coalition formation, constitutional constraints, administrative capacity and stakeholder negotiation cannot be collapsed into synthetic agent scores.

Research agenda: from pedagogical swarm to institutional-grade system

The notebook suggests a clear research agenda.

A first extension would create a versioned Mexican legal graph in which articles, fractions, definitions, authorities and cross-references are represented as explicit nodes and edges. Proposed language could then be checked against dependency structure before drafting proceeds.

A second extension would replace synthetic perspective alone with layered evidence. Real consultation submissions could enter as signed, provenance-preserving documents. Market data could quantify liquidity, issuance cost or enforcement frequency. Judicial decisions could be indexed as a separate authority class. The system would distinguish *what participants want*, *what the law says* and *what the data show*.

A third extension would introduce heterogeneous models. One model could propose, a second could attack legal coherence, and a third could search for unsupported factual assumptions. Cross-model disagreement would become another governance signal.

A fourth extension would formalize the human committee. Instead of a single Boolean authorization flag, a review interface could require named approvals for corpus adequacy, constitutional analysis, market impact, drafting technique and policy desirability. Human authorization would then be as structured as model participation.

A fifth extension would evaluate robustness across repeated runs. The research question would become: which article candidates remain stable under changes in seed, model, retrieval method, agent mix and scoring weights? Stability analysis could identify high-confidence agenda items while highlighting proposals that are artifacts of architecture choices.

Finally, the system could simulate consequences without pretending to predict them perfectly. Scenario engines might estimate how a proposed rule changes listing incentives, compliance burden, intermediary behavior or investor remedies under different assumptions. The legislative swarm would then interact with a separate economic-impact layer rather than asking the language model to narrate impacts from intuition.

Conclusion: the swarm expands thought; it does not inherit sovereignty

The simplest way to use a language model for legislative reform is to ask it to draft. The notebook rejects that shortcut. It converts legislative reasoning into a governed sequence: define the authoritative corpus, represent the law, create explicit institutional perspectives, diagnose independently, form temporary coalitions, generate candidate reforms, subject them to adversarial review, audit the process, rank article-level alternatives, require human authorization, draft only the selected provisions, integrate the package, validate mechanical integrity and export the entire genealogy.

The value of the approach is not that fifty agents become fifty legislators. They do not. The value is that the architecture makes more of the hidden work of reform visible. It turns interests into declared lenses, source authority into a corpus rule, criticism into a required data structure, selection into an inspectable function, and uncertainty into an exportable finding.

The stored execution reinforces the point. The pipeline completed. It produced a twenty-article package. It passed deterministic integrity checks. Yet nineteen legislative-technique reviews remained unresolved. A less governed system might hide that fact behind the fluency of the integrated initiative. This system preserves it and changes the final label to what the evidence supports: *anteproyecto para revisión humana*.

FINAL PROPOSITION

The relevant measure of success is not whether the initiative sounds professional or whether fifty synthetic agents agree. Success is whether an independent reviewer can reconstruct the corpus, the diagnoses, the disagreements, the selection rule, the drafts, the unresolved objections and the boundary of machine authority. The swarm expands the institution's capacity to think. The decision about what law should govern the market remains human and institutional.

Appendix: notebook-to-monograph map

Notebook	Function	Monograph interpretation
Cell 1	Configuration and dependencies	Reproducibility, credential discipline, centralized model access and audit structures.
Cell 2	Authorized corpus	Epistemic boundary: only designated legal sources receive authority.
Cell 3	Article segmentation	Deterministic legal identity and article-level retrieval.
Cell 4	Fifty-agent swarm	Institutional specialization through explicit roles, biases and success criteria.
Cell 5	Claude gateway	Bounded model access and structured output contracts.
Cell 6	Round 1 diagnosis	Independent problem discovery before solution generation.
Cell 7	Problem map and constellations	Deterministic synchronization and adaptive cross-family working groups.
Cell 8	Round 2 proposals	Structured reform hypotheses and article-coverage expansion.
Cell 9	Round 3 adversarial review	Independent counter-perspective scoring and preservation of objections.
Cell 10	Outer-loop audit	Governance of participation, concentration, duplication and process quality.
Cell 11	Multicriteria selection	Transparent article-level portfolio construction and human gate.
Cell 12	Legislative drafting	Article-specific drafting followed by independent technique review.
Cell 13	Integration	Coherent statement of reasons without expanding the authorized package.
Cell 14	Validation	Deterministic integrity checks and explicit unresolved legal work.
Cell 15	Export	Preservation of product, audit trail, governance report and validation record.

Appendix: source notebook and figure corpus

This monograph is based on the supplied Colab notebook `LMV_50_Agent_Swarm_Reform(2).ipynb`, available through the source notebook on Google Drive. The ten figures in the `figures/` directory were extracted directly from the notebook's embedded image cells and are included without redesign so that the monograph remains visually traceable to the instructional artifact.

The numerical results reported in this document are taken from the stored notebook outputs. They describe that execution, not an independently rerun experiment. In particular, the monograph preserves the notebook's own distinction between deterministic readiness for human review and substantive legislative-technique approval.

BUNDLE DESIGN FOR PRISM

The bundle is self-contained. Keep the main `.tex` file at the root and the supplied `figures/` directory beside it. No external image URLs are required. The preamble reproduces the visual language of the supplied reference monograph: A4 page geometry, navy/blue/teal palette, section rhythm, running heads, folios and square-edged callout boxes.

References

1. **Source notebook.** *LMV 50-Agent Swarm Reform*, supplied Colab notebook, October 2026. [Open the source notebook on Google Drive](#). All architecture descriptions, stored-run metrics, selected article numbers, validation results and notebook figures in this monograph derive from that source.
2. **Reference visual system.** *From Airspace to Markets: A Governed Autonomous Control Architecture*, supplied LaTeX monograph. The present bundle reproduces its typography, color palette, page rhythm, callout-box family, running-head structure and cover grammar.
3. **Authorized legal corpus.** The notebook is designed to ingest the Mexican *Ley del Mercado de Valores* and any additional user-authorized legal sources with provenance. This monograph reports the notebook methodology and does not independently certify the legal currency or doctrinal correctness of the corpus used in the stored run.

Status. Pedagogical research monograph. The underlying notebook output is an anteproyecto for mandatory human legal review and does not constitute a filed legislative initiative, a legal opinion or legal advice.